UNIVERSIDAD DEL VALLE DE GUATEMALA
Facultad de Ingeniería
Departamento de Ciencias de la Computación
CC3067 – Redes de Computadoras
Sección 10
Ing. Kevin Velásquez



Lab3 - Algoritmos de Enrutamiento

Paula Daniela de León Godoy – 23202
Nadissa Vela – 23754

GUATEMALA, 18 de octubre de 2026

Tablas de enrutamiento

Salida de show ip route de los cinco routers, después de configurar OSPF, EIGRP y la redistribución.

R-OSPF-1

```
R-OSPF-1>show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP

       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area

       * - candidate default, U - per-user static route, o - ODR

       P - periodic downloaded static route


Gateway of last resort is not set


     10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks

C       10.0.0.0/30 is directly connected, Serial0/0/0

L       10.0.0.1/32 is directly connected, Serial0/0/0

C       10.0.0.4/30 is directly connected, Serial0/0/1

L       10.0.0.5/32 is directly connected, Serial0/0/1

O       10.0.0.8/30 [110/128] via 10.0.0.6, 01:29:20, Serial0/0/1

     172.16.0.0/24 is subnetted, 1 subnets

O E1    172.16.0.0/24 [110/84] via 10.0.0.6, 01:22:40, Serial0/0/1

O    192.168.0.0/24 [110/129] via 10.0.0.2, 01:29:10, Serial0/0/0

                    [110/129] via 10.0.0.6, 01:29:10, Serial0/0/1

     192.168.1.0/24 is variably subnetted, 4 subnets, 3 masks

C       192.168.1.0/26 is directly connected, GigabitEthernet0/0.10

L       192.168.1.1/32 is directly connected, GigabitEthernet0/0.10

C       192.168.1.64/28 is directly connected, GigabitEthernet0/0.30

L       192.168.1.65/32 is directly connected, GigabitEthernet0/0.30
```

R-OSPF-2

```
R-OSPF-2>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/0/0
L       10.0.0.2/32 is directly connected, Serial0/0/0
O       10.0.0.4/30 [110/128] via 10.0.0.10, 01:29:34, Serial0/0/1
C       10.0.0.8/30 is directly connected, Serial0/0/1
L       10.0.0.9/32 is directly connected, Serial0/0/1
     172.16.0.0/24 is subnetted, 1 subnets
O E1    172.16.0.0/24 [110/84] via 10.0.0.10, 01:23:04, Serial0/0/1
     192.168.0.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.0.0/24 is directly connected, GigabitEthernet0/0
L       192.168.0.1/32 is directly connected, GigabitEthernet0/0
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
O       192.168.1.0/26 [110/129] via 10.0.0.1, 00:25:09, Serial0/0/0
                       [110/129] via 10.0.0.10, 00:25:09, Serial0/0/1
O       192.168.1.64/28 [110/129] via 10.0.0.1, 01:29:34, Serial0/0/0
                        [110/129] via 10.0.0.10, 01:29:34, Serial0/0/1
```

R-EIGRP-1

```
R-EIGRP-1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
```

```
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
D EX    10.0.0.0/30 [170/1433600] via 10.0.0.18, 01:15:36, Serial0/0/1
D EX    10.0.0.4/30 [170/1433600] via 10.0.0.18, 01:15:36, Serial0/0/1
D EX    10.0.0.8/30 [170/1433600] via 10.0.0.18, 01:15:36, Serial0/0/1
C       10.0.0.12/30 is directly connected, Serial0/0/0
L       10.0.0.13/32 is directly connected, Serial0/0/0
C       10.0.0.16/30 is directly connected, Serial0/0/1
L       10.0.0.17/32 is directly connected, Serial0/0/1
D       10.0.0.20/30 [90/1536000] via 10.0.0.18, 01:15:51, Serial0/0/1
                     [90/1664000] via 10.0.0.14, 01:31:33, Serial0/0/0
     172.16.0.0/16 is variably subnetted, 5 subnets, 4 masks
D       172.16.0.0/25 [90/514560] via 10.0.0.14, 01:38:45, Serial0/0/0
                      [90/1538560] via 10.0.0.18, 01:15:51, Serial0/0/1
C       172.16.0.128/26 is directly connected, GigabitEthernet0/0.10
L       172.16.0.129/32 is directly connected, GigabitEthernet0/0.10
C       172.16.0.192/27 is directly connected, GigabitEthernet0/0.20
L       172.16.0.193/32 is directly connected, GigabitEthernet0/0.20
D    192.168.0.0/23 [90/3321856] via 10.0.0.18, 01:15:36, Serial0/0/1
```

R-EIGRP-2

```
R-EIGRP-2>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

```
Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
D EX    10.0.0.0/30 [170/1433600] via 10.0.0.22, 01:15:52, Serial0/0/1
D EX    10.0.0.4/30 [170/1433600] via 10.0.0.22, 01:15:52, Serial0/0/1
D EX    10.0.0.8/30 [170/1433600] via 10.0.0.22, 01:15:52, Serial0/0/1
C       10.0.0.12/30 is directly connected, Serial0/0/0
L       10.0.0.14/32 is directly connected, Serial0/0/0
D       10.0.0.16/30 [90/1536000] via 10.0.0.22, 01:16:07, Serial0/0/1
                     [90/1664000] via 10.0.0.13, 01:34:44, Serial0/0/0
C       10.0.0.20/30 is directly connected, Serial0/0/1
L       10.0.0.21/32 is directly connected, Serial0/0/1
     172.16.0.0/16 is variably subnetted, 4 subnets, 4 masks
C       172.16.0.0/25 is directly connected, GigabitEthernet0/0
L       172.16.0.1/32 is directly connected, GigabitEthernet0/0
D       172.16.0.128/26 [90/514560] via 10.0.0.13, 01:39:01, Serial0/0/0
                        [90/1538560] via 10.0.0.22, 01:16:07, Serial0/0/1
D       172.16.0.192/27 [90/514560] via 10.0.0.13, 01:39:01, Serial0/0/0
                        [90/1538560] via 10.0.0.22, 01:16:07, Serial0/0/1
D    192.168.0.0/23 [90/3321856] via 10.0.0.22, 01:15:52, Serial0/0/1
```

R-CENTRAL

```
R-CENTRAL>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set
```

```
     10.0.0.0/8 is variably subnetted, 10 subnets, 2 masks
O       10.0.0.0/30 [110/192] via 10.0.0.5, 01:30:27, Serial0/0/0
                    [110/192] via 10.0.0.9, 01:30:27, Serial0/0/1
C       10.0.0.4/30 is directly connected, Serial0/0/0
L       10.0.0.6/32 is directly connected, Serial0/0/0
C       10.0.0.8/30 is directly connected, Serial0/0/1
L       10.0.0.10/32 is directly connected, Serial0/0/1
D       10.0.0.12/30 [90/640000] via 10.0.0.17, 01:17:15, Serial0/1/0
                     [90/640000] via 10.0.0.21, 01:16:23, Serial0/1/1
C       10.0.0.16/30 is directly connected, Serial0/1/0
L       10.0.0.18/32 is directly connected, Serial0/1/0
C       10.0.0.20/30 is directly connected, Serial0/1/1
L       10.0.0.22/32 is directly connected, Serial0/1/1
     172.16.0.0/16 is variably subnetted, 4 subnets, 4 masks
S       172.16.0.0/24 is directly connected, Null0
D       172.16.0.0/25 [90/386560] via 10.0.0.21, 01:16:23, Serial0/1/1
D       172.16.0.128/26 [90/386560] via 10.0.0.17, 01:17:15, Serial0/1/0
D       172.16.0.192/27 [90/386560] via 10.0.0.17, 01:17:15, Serial0/1/0
D    192.168.0.0/23 is a summary, 01:16:08, Null0
O    192.168.0.0/24 [110/65] via 10.0.0.9, 01:30:27, Serial0/0/1
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
O       192.168.1.0/26 [110/65] via 10.0.0.5, 00:25:56, Serial0/0/0
O       192.168.1.64/28 [110/65] via 10.0.0.5, 01:30:37, Serial0/0/0
```

Evidencia del balanceo OSPF para costo igual

En R-OSPF-1 el enlace directo hacia R-OSPF-2 siendo Serial0/0/0 tiene costo 128 y el enlace hacia R-CENTRAL siendo Serial0/0/1 tiene costo 64. El camino por R-CENTRAL cruza dos enlaces de costo 64, por lo que ambos caminos suman 128.

show ip ospf interface

```
R-OSPF-1>show ip ospf interface


GigabitEthernet0/0.30 is up, line protocol is up
  Internet address is 192.168.1.65/28, Area 0
  Process ID 1, Router ID 1.1.1.1, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State WAITING, Priority 1
  No designated router on this network
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    No Hellos (Passive interface)
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
Serial0/0/1 is up, line protocol is up
  Internet address is 10.0.0.5/30, Area 0
  Process ID 1, Router ID 1.1.1.1, Network Type POINT-TO-POINT, Cost: 64
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:05
  Index 2/2, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
```

```
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1 , Adjacent neighbor count is 1
    Adjacent with neighbor 3.3.3.3
  Suppress hello for 0 neighbor(s)
Serial0/0/0 is up, line protocol is up
  Internet address is 10.0.0.1/30, Area 0
  Process ID 1, Router ID 1.1.1.1, Network Type POINT-TO-POINT, Cost: 128
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:04
  Index 3/3, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1 , Adjacent neighbor count is 1
    Adjacent with neighbor 2.2.2.2
  Suppress hello for 0 neighbor(s)
GigabitEthernet0/0.10 is up, line protocol is up
  Internet address is 192.168.1.1/26, Area 0
  Process ID 1, Router ID 1.1.1.1, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 1.1.1.1, Interface address 192.168.1.1
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    No Hellos (Passive interface)
  Index 4/4, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
```

La ruta hacia la LAN de R-OSPF-2, al utilizar la misma métrica, se obtiene dos saltos distintos.

show ip route 192.168.0.0

```
R-OSPF-1>show ip route 192.168.0.0
Routing entry for 192.168.0.0/24
Known via "ospf 1", distance 110, metric 129, type intra area
  Last update from 10.0.0.2 on Serial0/0/0, 01:18:28 ago
  Routing Descriptor Blocks:
  * 10.0.0.2, from 2.2.2.2, 01:18:28 ago, via Serial0/0/0
      Route metric is 129, traffic share count is 1
    10.0.0.6, from 2.2.2.2, 01:18:28 ago, via Serial0/0/1
      Route metric is 129, traffic share count is 1
```

Evidencia del balanceo EIGRP de variance 3

En R-EIGRP-1 la red 172.16.0.0/25 se alcanza por el enlace directo de dirección 10.0.0.14 con métrica 514560 y por R-CENTRAL con dirección 10.0.0.18 con métrica 1538560. La segunda es fisible porque su distancia reportada de 386560 es menor que la distancia factible de 514560. La relación de métricas es de aproximadamente 3 a 1, siendo equivalente a 75% y 25% del tráfico.

show ip eigrp topology

```
R-EIGRP-1#show ip eigrp topology 172.16.0.0 255.255.255.128

IP-EIGRP (AS 100): Topology entry for 172.16.0.0/25

  State is Passive, Query origin flag is 1, 2 Successor(s), FD is 514560

  Routing Descriptor Blocks:

  10.0.0.14 (Serial0/0/0), from 10.0.0.14, Send flag is 0x0

      Composite metric is (514560/28160), Route is Internal

      Vector metric:

        Minimum bandwidth is 10000 Kbit

        Total delay is 10100 microseconds

        Reliability is 255/255

        Load is 1/255

        Minimum MTU is 1500

        Hop count is 1

  10.0.0.18 (Serial0/0/1), from 10.0.0.18, Send flag is 0x0

      Composite metric is (1538560/386560), Route is Internal

      Vector metric:

        Minimum bandwidth is 10000 Kbit

        Total delay is 50100 microseconds

        Reliability is 255/255

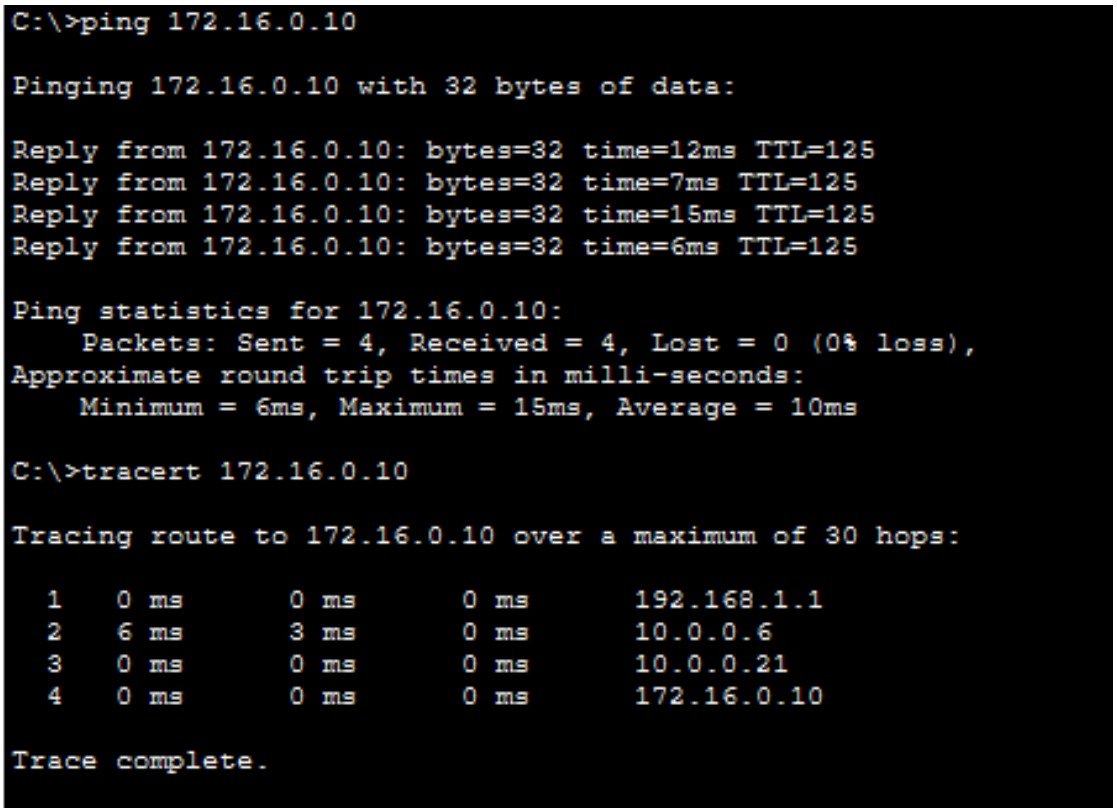
        Load is 1/255

        Minimum MTU is 1500

        Hop count is 2
```

Prueba de conectividad extremo a extremo

Ping y tracert de PC-O1-V10 siendo VLAN 10 del lado OSPF hacia PC-E2 siendo 172.16.0.10, lado EIGRP.



```
C:\>ping 172.16.0.10

Pinging 172.16.0.10 with 32 bytes of data:

Reply from 172.16.0.10: bytes=32 time=12ms TTL=125
Reply from 172.16.0.10: bytes=32 time=7ms TTL=125
Reply from 172.16.0.10: bytes=32 time=15ms TTL=125
Reply from 172.16.0.10: bytes=32 time=6ms TTL=125

Ping statistics for 172.16.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 6ms, Maximum = 15ms, Average = 10ms

C:\>tracert 172.16.0.10

Tracing route to 172.16.0.10 over a maximum of 30 hops:

  1    0 ms      0 ms      0 ms      192.168.1.1
  2    6 ms      3 ms      0 ms      10.0.0.6
  3    0 ms      0 ms      0 ms      10.0.0.21
  4    0 ms      0 ms      0 ms      172.16.0.10

Trace complete.
```